

 PYTHON PROGRAMMING CORE

File Handling in Python

A comprehensive guide to persistent data storage, file operations, access modes, and object serialization with corrected code examples.

Understanding File Storage

Why Files are Needed

Variables and data structures reside in Volatile Memory (RAM), which loses data once program execution terminates.

Files provide **Permanent Storage** on non-volatile media (HDD/SSD), allowing stored information to be accessed, processed, and preserved for future operations.

Fundamental File Types

Text Files: Store character data (ASCII/UTF-8).

Human-readable, structured row by row (e.g., .txt, .csv, .json).

Binary Files: Store raw bytes without character encoding.

Optimized for non-text media (e.g., .png, .mp4, .exe, .dat).

File Access Modes Reference

Mode	Description & Behavior	File Creation	Pointer Position
'r' 'r'	Read existing text file (Default). Raises FileNotFoundError if missing.	No	Start of file
'w' 'w'	Write mode. Overrides existing contents completely.	Yes	Start of file
'a' 'a'	Append mode. Retains existing data and appends at the end.	Yes	End of file
'r+' 'r+'	Read and Write data. Previous contents are preserved initially.	No	Start of file
'w+' 'w+'	Write and Read data. Overwrites existing contents immediately.	Yes	Start of file
'a+' 'a+'	Append and Read data without overwriting existing content.	Yes	End of file
'x' 'x'	Exclusive creation for writing. Fails with FileExistsError if file exists.	Yes	Start of file

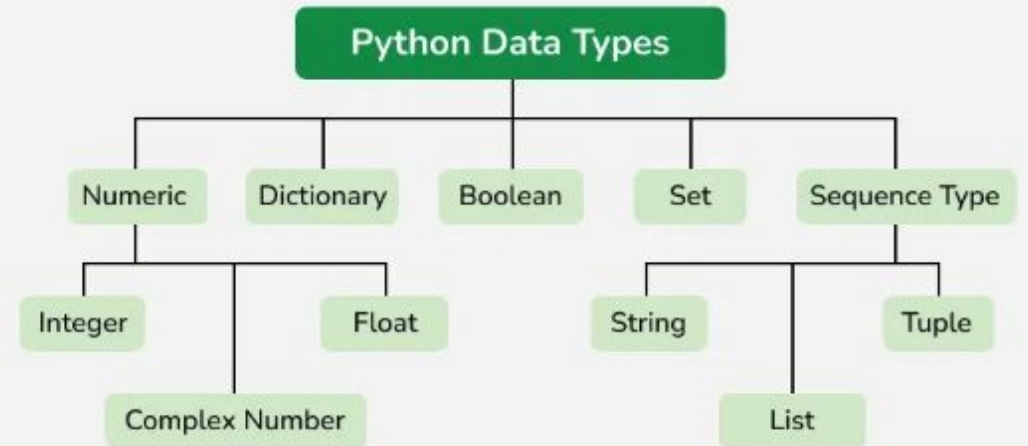
* Note: Appending 'b' to any mode enables Binary Mode (e.g., 'rb', 'wb', 'ab').

Opening & Closing Files Safely

Opening requires Python's built-in `open()` function. Closing releases system resources.

```
# Traditional Method
f = open("abc.txt", 'w')
# Perform operations ...
f.close()

# Best Practice: Context Manager (with)
with open("abc.txt", 'w') as f:
    f.write("Data saved automatically\n")
# File closes automatically even if exceptions occur
```



File Object Properties & Inspection

-  **name Property:** Returns the name or path of the opened file
-  **object mode Property:** Displays the access mode in which the file was opened.
-  **closed Property:** Returns boolean (True/False) indicating whether the file is closed.
-  **readable() / writable():** Methods returning booleans checking if read or write operations are permitted.

```
# Inspection Example (Corrected Code)
f = open("abc.txt", 'w')
print("File Name:", f.name)
print("File Mode:", f.mode)
print("Is Readable:", f.readable())
print("Is Writable:", f.writable())
f.close()
print("Is Closed:", f.closed)
```

```
File Name: abc.txt
File Mode: w
Is Readable: False
Is Writable: True
Is Closed: True
```

Reading & Writing Operations



Writing Methods

write(str): Writes a single string to the file.

writelines(list): Writes an iterable sequence/list of strings without automatic newlines.



Reading Methods

read([n]): Reads entire content, or up to n characters.

readline(): Reads a single line ending in \n.

readlines(): Reads all lines into a list of strings.



Execution Code

```
with open("abc.txt", 'w') as f:
    f.write("Durga\n")
    f.writelines(["Gitam\n", "College\n"])

with open("abc.txt", 'r') as f:
    lines = f.readlines()
```

Working with Binary Files

Binary Operations (rb / wb)

Binary mode processes raw bytes rather than text strings, crucial for media handling like images, audio, and compressed archives.

```
# Copying Image File
with open("flower.jpg", "rb") as f1:
    data = f1.read()

with open("newpic.jpg", "wb") as f2:
    f2.write(data)

print("Binary copy created successfully!")
```



Object Serialization with Pickle

Pickling converts Python objects (lists, dicts) into byte streams for storage; Unpickling restores them.

```
import pickle

data = ["Item 1", 100, 99.5]

# Serialization (Dump)
with open("pickle_file.dat", "wb") as f:
    pickle.dump(data, f)

# Deserialization (Load)
with open("pickle_file.dat", "rb") as f:
    loaded_data = pickle.load(f)
print(loaded_data) # ['Item 1', 100, 99.5]
```

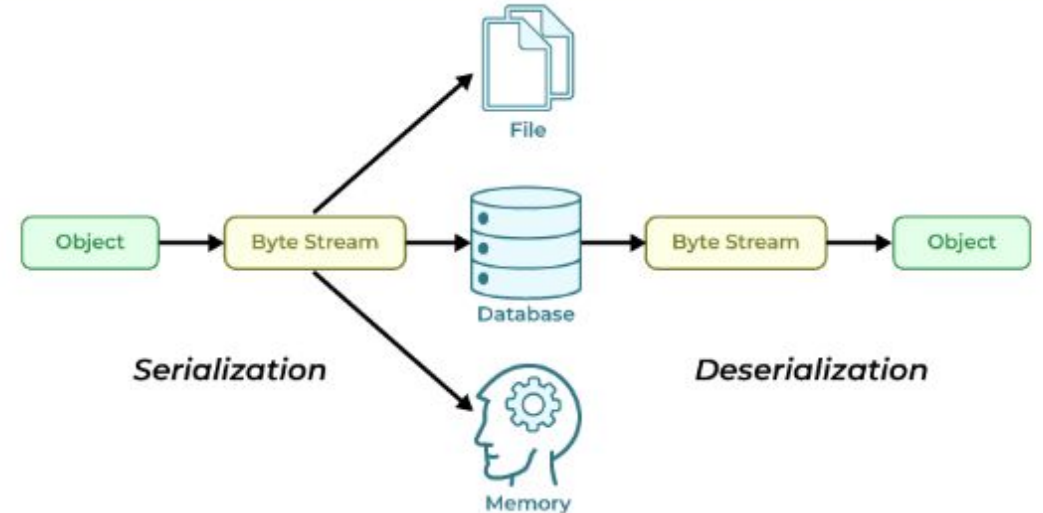


Image Sources



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